

HOST

Pinned Expert Store

Host DRAM



Control Plane

Dynamic & Uncaptured

- Route computation
- Cache policy
- Wave planning
- Slot assignment

Logical → Physical Mapping

Core Abstraction

Logical ID	Slot	Version
e	S	v
e	S	v
e	S	v

NPU / HBM (Captured Graph Boundary)

Captured Router

Router Operation

Top-k Selection

Input: Hidden states

Output: Expert IDs + Weights

Example: $[e, e, e]$

Captured (stable address)
Native routing logic
No approximation

IDs

Staging Boundary (Uncaptured)

Cache Check Wave Planning

For each routed expert:

- Cache hit → Keep slot
- Cache miss →
Assign new slot
H2D transfer
Version++

Dynamic decisions
Capacity-bounded

Map

Captured Computation

Fixed Slot Bank

Replay-stable addresses



Stable refs

Fused MoE Kernel

Grouped MatMul + Combine

Reads from stable slot addresses

Graph-captured
Fixed addresses

Key Insight: Logical ownership changes (), but graph-visible storage () remains fixed